Shenqi Zhou

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Education

Zhejiang University Sep. 2023 - Jun. 2027 (expected)

Bachelor of Engineering in Biomedical Engineering

- GPA: **4.11/4.3** | Average: **91.12/100**

Research Interests

My research interests focus on AI for healthcare and the evaluation of LLMs and agents. My current project, Auto-MD-bench, studies whether a single performance metric is sufficient to guide autoresearch agents toward models for real-world medical use.

Manuscript Under Review

Shenqi Zhou[*], Qiran Zou[*], Wenhao Zhao, Zhenyu Liao, and Dianbo Liu. *Auto-MD-bench: Evaluating Autoresearch Agents for Real-World Medical Tasks.* Under review. arXiv:2610.25069. [*]Equal contribution.

Research Experience

Auto-MD-bench: Evaluating Autoresearch Agents for Real-World Medical Tasks

Research Intern, National University of Singapore | Adviser: Prof. Dianbo Liu *Aug. 2026 - Present*

- Led the design and implementation of a benchmark evaluating autoresearch agents beyond predictive accuracy: selected 10 clinical datasets and defined 15 survival, classification, and sequential prediction tasks, including five external validation tasks.
- Conducted all experiments and analyses comparing four agents using the same data, language model, and compute budget; assessed predictive performance, calibration, and generalization, and wrote the manuscript.
- Found that stronger predictive performance did not ensure well-calibrated risk estimates; analyzed agents' code and development records to identify modeling practices associated with this gap.
- Investigated whether changing agents' optimization objectives could address this gap; demonstrated improved calibration while maintaining predictive performance in a follow-up experiment.

Evaluating Chinese Sentence Understanding in Large Language Models

Undergraduate Researcher, Zhejiang University | Adviser: Prof. Nai Ding *Apr. 2025 - Mar. 2026*

- Adapted Sentence-Space Metrics to compare human and LLM judgments of semantic relatedness and entailment direction in Chinese.
- Constructed 370 sentences from 50 premises and evaluated ordered pairs with three LLMs; collected human judgments on 2,577 selected pairs and analyzed sparsity, clustering, and representational similarity.
- Found closer human-model agreement for semantic relatedness than for entailment direction, highlighting a gap between recognizing shared content and judging directional inference.

Relevant Coursework

Mathematics and statistics: Probability and Mathematical Statistics (94); Linear Algebra (A) (93); Calculus (A) II (93); Complex Variable Functions & Integral Transformation (92); Error Theory and Data Processing (99).
Computing: Advanced Programming (100); Python Programming (H) (98); Lab & Fundamentals of C Programming (95); Introduction to Pattern Analysis and Machine Learning (93); Data Structures and Algorithm Analysis (90).
Biomedical engineering: Biomedical Signal Processing (96); Biomedical Imaging Technology (95); Engineering Physiology (92).

Honors and Awards

Zhejiang University Second-Class Scholarship | 2023-2024, 2024-2025
Bioer Scholarship, Zhejiang University | 2024-2025

Skills and Languages

Programming: Python, C.
AI tools: Proficient with coding agents.
Languages: Chinese (native); English (IELTS Academic: 7.0/9, July 2026).